Statistical Supervised Machine Learning

TP 4 – From an error on a sample to a PAC-Bayesian guarantee

Master MIASHS & MALIA

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Abstract

Fourth of five practical sessions (1: protocol, 2: bagging, 3: boosting, 4 and 5: PAC-Bayes). This session follows Sections 13 and 14 of the lecture notes, step by step. On a toy problem in which every true risk is known exactly, so that we can draw thousands of samples and *count* how often a bound fails, we (1) certify one classifier fixed in advance with Hoeffding's inequality and with the sharper kl bound, (2) see what goes wrong when the classifier is chosen with the sample and repair it with the union bound, (3) spread the confidence over the classifiers with a *prior* (Occam's bound), and then move from one classifier to a *distribution* over classifiers: (4) the KL divergence, (5) the first PAC-Bayesian bound of the notes, with its parameter λ, and the bounds of McAllester and Seeger used in practice, (6) the Gibbs posterior, and (7) a check that all these bounds hold. Majority votes are the subject of TP 5. The session lasts **three hours**; Part 8 contains the difficult questions.

About this session – TP 4: From an error on a sample to a PAC-Bayesian guarantee

Goal. By the end of the session you should be able to compute the test-set and kl bounds of a fixed classifier, explain why a selected classifier needs a union bound and how a prior spreads the confidence, compute the Gibbs risk and the KL divergence of a posterior, compute the PAC-Bayesian bound with parameter λ and the bounds of McAllester and Seeger, explain the trade-off made by the Gibbs posterior, and check by simulation that the bounds hold.
To reread before the session. Section 13 (from an error on a sample to a guarantee: Corollary 13.3, Theorem 13.5, Algorithm 12, Theorems 13.7 and 13.8) and Section 14 (the PAC-Bayesian bound: Definition 14.1, Theorems 14.4 and 14.5, Proposition 14.6) of the lecture notes. The exercises of Sections 13 and 14 of the exercise booklet are a good preparation.

Timing (3 hours).	Setting up and Part 1	35 min
	Part 2	15 min
	Part 3	20 min
	Part 4	20 min
	Part 5	35 min
	Part 6	30 min
	Part 7	25 min
	Part 8 ($\star$)	at home

To hand in. Your notebook with the tables of Parts 1 to 7, the figures of Parts 1, 5 and 6, and your proofs.
Tools. `numpy` and `matplotlib` (and the helpers of `common/tptools.py`).

The toy problem. The input X is uniform on $[0,1]$. The clean label is $\text{sign}(X-\theta^\star)$ with $\theta^\star = 0.37$, and it is flipped with probability $\eta = 0.15$. The class $\mathcal{H}$ contains the $K = 102$ threshold classifiers $h_{t,s}(x) = s \cdot \text{sign}(x-t)$, $t \in \{0, 0.02, \dots, 1\}$, $s \in \{-1, +1\}$. Unless stated otherwise, $\delta = 0.05$.

1. Show that $R_{\mathcal{D}}(h_{t,+1}) = \eta + (1-2\eta)|t-\theta^\star|$ and $R_{\mathcal{D}}(h_{t,-1}) = 1 - R_{\mathcal{D}}(h_{t,+1})$. What is the best classifier of $\mathcal{H}$?
2. Write `sample(m, rng)` and a *vectorized* function `emp_risks(x, y)` returning the 102 empirical risks (build the $102 \times m$ matrix of predictions).

1 One classifier fixed in advance: Hoeffding and kl bounds (Sections 13.1–13.4)

Fix $h = h_{0.6,+1}$, whose risk is $p = R_{\mathcal{D}}(h) = 0.311$. For a sample of size m, $m\widehat{R}_S(h) \sim \mathcal{B}(m,p)$: you can simulate $\widehat{R}_S(h)$ directly with `rng.binomial`.

1. **Test-set bound.** From Hoeffding's inequality (Theorem 13.2), $\mathbb{P}(R_{\mathcal{D}}(h) - \widehat{R}_S(h) \geq \varepsilon) \leq e^{-2m\varepsilon^2}$, prove Corollary 13.3: with probability at least $1-\delta$, $R_{\mathcal{D}}(h) \leq \widehat{R}_S(h) + \sqrt{\ln(1/\delta)/(2m)}$.
   Why must h be fixed before the sample is drawn?
2. **kl bound.** Theorem 13.5 gives, with probability at least $1-\delta$, $R_{\mathcal{D}}(h) \leq \overline{\text{kl}}^{-1}(\widehat{R}_S(h), \ln(1/\delta)/m)$, where $\overline{\text{kl}}^{-1}(q,\varepsilon) = \max\{p \in [q,1] : \text{kl}(q\|p) \leq \varepsilon\}$ and $\text{kl}(q\|p) = q\ln\frac{q}{p} + (1-q)\ln\frac{1-q}{1-p}$ (Definition 13.4).
   Implement $\overline{\text{kl}}^{-1}$ by bisection (Algorithm 12), for whole arrays of q at once, and check it against `kl_inv_upper` of `common/tptools.py`; check that $\overline{\text{kl}}^{-1}(0.1, 0.05) \approx 0.220$ (example of Section 13.4).
   Using Pinsker's inequality $\text{kl}(q\|p) \geq 2(q-p)^2$, show that the kl bound is never worse than the test-set bound.
3. For $m \in \{50, 100, 500, 2000\}$, draw 10 000 samples and compute the mean of each bound and its *coverage* (the proportion of samples for which $R_{\mathcal{D}}(h)$ is below the bound).
   Comment.

4. For $m = 100$, plot the width "bound $-\widehat{R}_S(h)$" of both bounds as a function of the observed error $\widehat{R}_S(h) \in [0, 0.5]$.
   When is the kl bound much better?

2 What goes wrong when we choose: the union bound (Section 13.5)

1. Using the union bound (Lemma 13.6), prove Theorem 13.7: with probability at least $1-\delta$, *for all* $k \leq K$ simultaneously, $R_{\mathcal{D}}(h_k) \leq \widehat{R}_S(h_k) + \sqrt{\ln(K/\delta)/(2m)}$; give its kl version $R_{\mathcal{D}}(h_k) \leq \overline{\mathrm{kl}}^{-1}(\widehat{R}_S(h_k), \ln(K/\delta)/m)$.
   Why does it hold for a classifier *selected* on the sample, whatever the selection rule?

2. For $m \in \{50, 100, 400\}$ and 5000 samples, select the empirical risk minimizer (ERM) $\hat{h}$ among the 102 thresholds.
   Compute the mean of $R_{\mathcal{D}}(\hat{h}) - \widehat{R}_S(\hat{h})$, and the coverage of (a) the test-set bound with δ (as if $\hat{h}$ were fixed), (b) the union bound for $\hat{h}$, (c) the union bound for *all* the classifiers at once.
   Comment.

3 Occam's bound: spreading the confidence with a prior (Section 13.6)

Let P be a probability distribution on $\mathcal{H}$ (the *prior*), chosen *before* seeing the data. The kl version of Occam's bound (Theorem 13.8) states that, with probability at least $1-\delta$,

$$\forall h \in \mathcal{H}, \qquad R_{\mathcal{D}}(h) \leq \overline{\mathrm{kl}}^{-1}\Big(\widehat{R}_S(h),\ \frac{\ln \frac{1}{P(h)} + \ln \frac{1}{\delta}}{m}\Big).$$

1. Prove it, by applying the kl bound of Part 1 to each h with the confidence $\delta P(h)$ and the union bound.
   What do you recover with the uniform prior?

2. On the toy problem ($m = 100$, 2000 samples), compare three priors: uniform; "informative", with weights proportional to $e^{-(t-0.4)^2/(2\times 0.1^2)}$ on the threshold and 90% of the mass on $s = +1$; "misleading", the same centred on 0.9.
   For each prior, report the mean bound and the mean true risk of the ERM and of the classifier that minimizes the *bound*, and the coverage for all h at once.

3. Can you choose the prior after looking at the data?
   What happens if the prior is bad?

From one classifier to a distribution over classifiers. We keep the toy problem and its class $\mathcal{H}$ of $K = 102$ thresholds; the prior P is now uniform on $\mathcal{H}$. A *posterior* is any distribution Q on $\mathcal{H}$, possibly chosen using the sample S of size m. We recall (Sections 12 and 14 of the lecture notes):

- the **Gibbs classifier** G_Q draws $h \sim Q$ and predicts $h(x)$; its risks are $R_{\mathcal{D}}(G_Q) = \sum_h Q(h) R_{\mathcal{D}}(h)$ and $\widehat{R}_S(G_Q) = \sum_h Q(h) \widehat{R}_S(h)$ (Definition 12.1); the **majority vote** B_Q is the subject of TP 5;
- the **KL divergence** $\mathrm{KL}(Q\|P) = \sum_h Q(h) \ln \frac{Q(h)}{P(h)}$, with $0 \ln 0 = 0$ (Definition 14.1);
- the **PAC-Bayesian bound** of Theorem 14.4: for $\lambda > 0$ *fixed before seeing* S, with probability at least $1-\delta$, for all Q simultaneously,

$$R_{\mathcal{D}}(G_Q) \leq \widehat{R}_S(G_Q) + \frac{\mathrm{KL}(Q\|P) + \ln \frac{1}{\delta}}{\lambda} + \frac{\lambda}{8m};$$

- the bounds of **McAllester** and **Seeger** (Theorem 14.5): for $m \geq 8$, with probability at least $1-\delta$, for all Q simultaneously,

$$\text{(McAllester)} \quad R_{\mathcal{D}}(G_Q) \leq \widehat{R}_S(G_Q) + \sqrt{\frac{\mathrm{KL}(Q\|P) + \ln \frac{2\sqrt{m}}{\delta}}{2m}},$$

$$\text{(Seeger)} \quad R_{\mathcal{D}}(G_Q) \leq \overline{\mathrm{kl}}^{-1}\Big(\widehat{R}_S(G_Q),\ \frac{\mathrm{KL}(Q\|P) + \ln \frac{2\sqrt{m}}{\delta}}{m}\Big).$$

> **Good practice** Work with *vectors*: a posterior is an array of 102 nonnegative numbers summing to 1, so that $\widehat{R}_S(G_Q)$ is `Q @ r` and $R_{\mathcal{D}}(G_Q)$ is `Q @ R`. To compute $\mathrm{KL}(Q\|P)$, sum only over the h with $Q(h) > 0$. To compute weights $\propto e^{a_h}$ for large a_h, subtract $\max_h a_h$ before taking the exponential.

4 Posteriors, Gibbs risk and KL divergence (Sections 12.2 and 14.1)

1. Show that, for the uniform prior, $\mathrm{KL}(Q\|P) = \ln K - H(Q)$ with $H(Q) = -\sum_h Q(h) \ln Q(h) \geq 0$, so that $0 \leq \mathrm{KL}(Q\|P) \leq \ln K$.
   What is the KL of the uniform distribution on k of the K classifiers (Example 14.1)? Of a Dirac mass?
2. Draw one sample of size $m = 100$.
   For the following posteriors, compute $\widehat{R}_S(G_Q)$, $\mathrm{KL}(Q\|P)$ and the true Gibbs risk $R_{\mathcal{D}}(G_Q)$: (a) the Dirac mass on the ERM $\hat{h} = h_{\hat{t},\hat{s}}$; (b) the prior itself; (c) the uniform distribution on the 5 classifiers with the smallest empirical risks; (d) the uniform distribution on the $h_{t,+1}$ with $|t - \hat{t}| \leq 0.1$.
   Which posterior would you like to certify? What makes it "complex"?

5 The PAC-Bayesian bounds (Sections 14.3–14.5)

1. On the sample of Part 4, compute the bound of Theorem 14.4 for the four posteriors and $\lambda \in \{10, 30, 100, 300\}$.
   Plot it as a function of $\lambda \in [1, 10^4]$ for the posteriors (a) and (d).
   Which term dominates for a small λ, for a large λ?
2. For a fixed Q, show that the value of λ that minimizes the bound is $\lambda^\star = \sqrt{8m\big(\mathrm{KL}(Q\|P) + \ln\frac{1}{\delta}\big)}$, and that the minimum is $\widehat{R}_S(G_Q) + \sqrt{\big(\mathrm{KL}(Q\|P) + \ln\frac{1}{\delta}\big)/(2m)}$.
   Why can we *not* use $\lambda^\star$, computed from the data, in Theorem 14.4?
   Show that choosing λ in a grid of k values fixed in advance is allowed if $\ln\frac{1}{\delta}$ is replaced by $\ln\frac{k}{\delta}$, and compute this certificate for the four posteriors with the grid $\{10, 30, 100, 300\}$.
3. Compute McAllester's and Seeger's bounds for the four posteriors, and compare all the certificates.
   Compare the certificate of the Dirac mass with Occam's bound for $\hat{h}$ (Part 3).
   Why is the PAC-Bayesian bound a little looser here?

6 The Gibbs posterior (Section 14.6)

For $\lambda > 0$, the *Gibbs posterior* is $Q_\lambda(h) = \frac{P(h)\, e^{-\lambda \widehat{R}_S(h)}}{Z_\lambda}$ (Proposition 14.6): it minimizes $\widehat{R}_S(G_Q) + \frac{\mathrm{KL}(Q\|P)}{\lambda}$, hence the bound of Theorem 14.4 for this λ.

1. What is Q_λ for $\lambda = 0$ (where the formula still makes sense), and what is the limit of Q_λ when $\lambda \to \infty$?
2. On the sample of Part 4, for $\lambda \in \{0\} \cup [1, 10^4]$ (logarithmic grid of 41 values), plot $\widehat{R}_S(G_{Q_\lambda})$, $\mathrm{KL}(Q_\lambda\|P)/m$, Seeger's bound and $R_{\mathcal{D}}(G_{Q_\lambda})$ against λ.
   Plot Q_λ for a few values of λ.
3. Select λ by minimizing Seeger's bound.
   Why is Seeger's bound still valid for the selected posterior, without any union bound over the grid (compare with Part 2 and with question 2 of Part 5)?

7 Do the bounds hold?

1. For $m \in \{50, 100, 400\}$, draw 2000 samples.
   On each, select λ by minimizing Seeger's bound over the grid of Part 6, and record $\widehat{R}_S(G_{Q_\lambda})$, $R_{\mathcal{D}}(G_{Q_\lambda})$, Seeger's bound and whether it holds; record also the bound of Theorem 14.4 for the same posterior, with the union bound over the 41 positive values of the grid.
   Compare with Occam's bound for the ERM and its true risk.
2. Comment on the coverage, on the tightness of the bounds, and on the comparison with the ERM.

8 Going further ($\star$)

1. **The KL divergence is nonnegative.** Using Jensen's inequality for the concave function ln, prove that $\mathrm{KL}(Q\|P) \geq 0$, with equality if and only if $Q = P$ (Proposition 14.2).

2. **The Gibbs posterior is optimal.** Show that, for $\lambda > 0$, $\lambda \widehat{R}_S(G_Q) + \mathrm{KL}(Q\|P) = \mathrm{KL}(Q\|Q_\lambda) - \ln Z_\lambda$, where $Z_\lambda = \sum_h P(h) e^{-\lambda \widehat{R}_S(h)}$.
   Deduce Proposition 14.6.

3. **Proof of Theorem 14.4.** Let $\Delta(h) = R_{\mathcal{D}}(h) - \widehat{R}_S(h)$.
   (a) Using Hoeffding's lemma, $\mathbb{E}_S\left[e^{\lambda\Delta(h)}\right] \leq e^{\lambda^2/(8m)}$ for each fixed h, show that $\mathbb{E}_S \mathbb{E}_{h\sim P}\left[e^{\lambda\Delta(h)}\right] \leq e^{\lambda^2/(8m)}$.
   (b) Deduce with Markov's inequality (Lemma 13.1) that, with probability at least $1-\delta$, $\mathbb{E}_{h\sim P}\left[e^{\lambda\Delta(h)}\right] \leq \frac{1}{\delta} e^{\lambda^2/(8m)}$.
   (c) Conclude with the change of measure (Lemma 14.3), $\mathbb{E}_{h\sim Q}[\phi(h)] \leq \mathrm{KL}(Q\|P) + \ln \mathbb{E}_{h\sim P}\left[e^{\phi(h)}\right]$, applied to $\phi = \lambda\Delta$.
   Why does the result hold for all Q simultaneously, and why must λ be fixed in advance?